Lecture plan · for the lecturer

# Teaching cyber security at university: unit check

Cyber security · University · Unit check

| Time | Materials |
| --- | --- |
| 40 min | Projector or board, and the lecture slides. |

## Learning goals

- Principles and STRIDE
- Public-key cryptography
- Password storage and MFA
- Ethics and assessment

## Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

## Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change CTF assessment?

## Exit ticket

- One change to my security course this term, and how I will know it worked.

**Try it in your next lecture**
Bring your rules of engagement and CTF rubric to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

**1** **Storage defaulted to public and nobody changed it. The principle broken is**

◯ Fail-safe defaults ◯ Open design ◯ Psychological acceptability

**2** **Match each STRIDE threat to the property it violates.**

| | | |
|---|---|---|
| Spoofing | • • • • | Availability |
| Information disclosure | • • • • | Confidentiality |
| Denial of service | • • • • | Authentication |

**3** **Compute a public value.**

With p = 23, g = 5 and b = 2, Bob sends B = _____ .

**Word bank:** 2 · 10 · 25

**4** **Encrypting a message with Bob's public key proves who sent it.**

◯ True ◯ False

**5** **Plain Diffie–Hellman is vulnerable to**

◯ Man-in-the-middle ◯ Rainbow tables ◯ Collisions